ENERGY AND THE ENVIRONMENT

Climate Change – The Science

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- The greenhouse effect occurs naturally, providing a habitable climate.
- Atmospheric concentrations of some of the gases that produce the greenhouse effect are increasing due to human activity and most of the world's climate scientists consider that this is a significant part of the cause of observed climate change.
- The oceans are a critical part of the climate system, with vastly greater thermal capacity than the atmosphere. Most of the net energy increase in the climate system in recent decades is stored in the oceans. A small, slow increase in their temperature is significant.
- About one-third of human-induced greenhouse gas emissions come from the burning of fossil fuel to generate electricity. Nuclear power plants do not emit these gases.

The 'greenhouse effect' is the term used to describe the retention of heat in the Earth's lower atmosphere (troposphere) due to concentrations of certain trace gases and water vapour in the atmosphere. These gases are generally known as greenhouse gases (or more specifically as radiative gases). Concentrations of some of them have increased steadily during the 20th century and into the 21st, with carbon dioxide (CO_2) rising from under 300 parts per million (ppm) to over 400 ppm. A large part of the increase in all

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greenhouse gases is attributed to human sources, *i.e.* it is anthropogenic, hence the term 'anthropogenic global warming'.

Furthermore, although most sources of anthropogenic emissions can be identified in particular countries, their effect is in no way confined to those countries – it is global.

The greenhouse effect

The greenhouse effect itself occurs when short-wave solar radiation (which is not impeded by the greenhouse gases) heats the surface of the Earth, and the energy is radiated back through the Earth's atmosphere as heat, with a longer wavelength. In the wavelengths 5-30 μm a lot of this thermal radiation is absorbed by water vapour and carbon dioxide, which in turn radiate it, thus heating the atmosphere and land and ocean surface. This is natural and what keeps the Earth habitable. Without the greenhouse effect overnight temperatures would plunge and the average surface temperature would be about minus 18 °C, about the same as on the moon, which lacks the shroud of our atmosphere. We owe the difference of some 33 °C substantially to natural levels of water vapour (60%, or more including clouds) and carbon dioxide in the Earth's atmosphere.

In respect to enhancing the greenhouse effect, or the likelihood of AGW, the particular issue is focused in the 8-18 μm band where water vapour is a weak absorber of radiation and where the Earth's thermal radiation is greatest.* Increased concentrations of CO_2 and other radiative gases here mean that less heat is lost to space from the Earth's lower atmosphere, and temperatures at the Earth's surface are therefore likely to increase. Atmosphere and oceans are the focus of attention.

* Part of this 'window' (12.5-18 μm) is largely blocked by carbon dioxide absorption, even at the low levels originally existing in the atmosphere. The remainder of the 'window' coincides with the absorption proclivities of the other radiative gases: methane, (tropospheric) ozone, CFCs and nitrous oxide. It also appears that increased levels of carbon dioxide will increase the capture of heat in its main absorption band to some extent, though diminishing as levels increase, while more energy is absorbed in the weaker bands.
As well as the band consideration, methane is a stronger greenhouse gas because it has more atoms in the molecule than CO_2. The radiative effect is caused by infrared absorption, and molecules with more atoms absorb more infrared energy. IR absorption is by the electrons that bond between atoms in

a molecule and the way those atoms vibrate. More bonds = more vibrations = more IR absorption. Diatomic molecules, like O_2 and N_2 which mostly make up our atmosphere, absorb very little IR. CO_2, with two bonds, absorbs some IR, but it is the next most abundant gas so its effect is significant.

A number of indicators suggest that atmospheric warming due to increased levels of greenhouse gases is indeed observable since 1970, despite some masking by aerosols (see below). Global air temperatures do appear to have risen about 1.1 °C over the last 120-170 years, though this has been irregular rather than steady, and does not correlate well with the steady increase in greenhouse gas – notably CO_2 – concentrations. The amount, extent and rate of this exceeds natural climate variability, some of the warmest years on record have been in the last decade. However, the climate is a complex system and other factors influence global temperatures.

One of these is water vapour, and climate models have assumed that the direct warming effect of CO_2 is amplified by water vapour. However, there is doubt about whether in practice this occurs to the extent previously thought.

The oceans have also warmed slightly, affecting the climate.

Balancing factors

The major role of water vapour in absorbing thermal radiation is in some respects balanced by the fact that when condensed it causes an albedo effect which reflects about one-third of the incoming sunlight back into space. This effect is enhanced by atmospheric sulfate aerosols and dust, which provide condensation nuclei. Nearly half the sulfates in the atmosphere originate from sulfur dioxide emissions from power stations and industry, particularly in the northern hemisphere. Emissions of sulfates are increasingly constrained in most countries. Global sulphate emissions peaked in the early 1970s and decreased until 2000, with an increase since due mainly to increased emissions in China and from international shipping.

Volcanoes have contributed substantially to dust and acid aerosol levels high in the atmosphere. The Mount Pinatubo eruption in 1991 in the Philippines reduced average temperatures about half a degree Celsius (°C). While at lower levels in the atmosphere sulfate aerosols and dust are short-lived, such material in the stratosphere remains for years, increasing the amount of sunlight which is reflected away. Hence there is, for the time being, a balancing cooling effect on the Earth's surface. In the northern hemisphere the

sulfate aerosols are estimated to counter nearly half the heating effect due to anthropogenic greenhouse gases.

However, in many countries there are now programmes to reduce sulfur dioxide emissions from power stations, as these emissions cause acid rain. Hence this balancing factor will diminish and the rate of temperature increase due to greenhouse gases may consequently increase.

Global warming and climate change

There is clear evidence of changes in the composition of the greenhouse gases in the lower atmosphere, with CO_2 in particular steadily increasing to its present level of about 417 ppm (September 2021). This is one-and-a-half times its pre-industrial level. In May 2013 the daily mean concentration of carbon dioxide in the atmosphere of Mauna Loa, Hawaii, the primary global benchmark site, surpassed 400 ppm for the first time since measurements began there in 1958. It has increased by one-third in the last 200 years, and half of that in the last 30 years. In 2018 it rose 2.3 ppm (0.8%), and about 3 ppm in 2019 – the largest annual increase yet observed. Since then it has risen about 2.5 ppm per year. Ice core samples show that both carbon dioxide and methane levels are higher than at any time in the past 650,000 years – CO_2 generally being around 170-270 ppm up to the 20th century*.

* Carbon dioxide is essential to plant life, and needs to be at least 150 ppm to sustain it. At higher levels, plant growth is enhanced – the carbon dioxide fertilization effect. This removes about one-quarter of anthropogenic emissions and is responsible for much of the increase in photosynthesis worldwide since about 1900. Carbon dioxide cannot sensibly be called 'pollution' at any envisaged atmospheric levels.

Estimates of the individual contribution of particular gases to the greenhouse effect – their global warming potential (GWP), are broadly agreed (relative to carbon dioxide = 1). Such estimates depend on the physical behaviour of each kind of molecule and its lifetime in the atmosphere, as well as the gas's concentration*. Both direct and indirect effects due to interaction with other gases and radicals must be taken into account and some of the latter remain uncertain.

* Methane is 262% and nitrous oxide is 123% of the levels in 1750 according to the World Meteorological Organization.

Greenhouse gas	Concentration change, 1800s - 2018	Anthropogenic sources	100-yr GWP*	Proportion of total effect apart from water vapour (approximate)
carbon dioxide	280 - 408 ppm	fossil fuel burning, deforestation	1	66%
methane	0.75 - 1.87 ppm	agriculture, fuel leakage	25	17%
halocarbons	0 - 0.7 ppb	refrigerants	1100-11,000	11%
nitrous oxide	270 - 331 ppb	agriculture, combustion	300	6%
ozone	15? - 20-30 ppb	urban pollution		

* World Meteorological Organization, WMO Greenhouse Gas Bulletin No. 15 (25 November 2019)

In addition to these well-documented radiative gases there is increasing concern about sulfur hexafluoride (SF_6) used in grid switchgear, with about 8000 tonnes emitted annually and increased use envisaged. Its GWP is 23,900.

Considering three long-lived radiative gases closely linked to human activities – CO_2, CH_4 & N_2O – and their individual GWP, a figure in CO_2-equivalent can be expressed. In 2018 this reached 496 ppm according to the US National Oceanic and Atmospheric Administration (NOAA) Annual Greenhouse Gas Index (AGGI). From 1990 to 2023 there was a 51% increase in total radiative forcing, with CO_2 accounting for about 80% of this, according to figures from the NOAA, which is focused on the many sources, sinks and chemical transformations in the atmosphere.

Although water vapour has a major influence on absorbing long-wave thermal radiation, its GWP is not calculated since its concentration in the atmosphere varies widely and mainly depends on air temperature. Also its residence time is only about nine days, compared with years for CO_2 and methane. It is classed as a positive feedback, not a forcing agent for the troposphere. In the stratosphere, water vapour from methane oxidation and possibly from aircraft may be a forcing agent, but the former is included in methane's GWP.

The Intergovernmental Panel on Climate Change (IPCC) is a scientific body under the auspices of the UN, set up in 1988 to

review and assess scientific and other information on human contributions to climate change. It was set up as a partnership between the World Meteorological Organisation (WMO) and the UN Environment Program (UNEP) and 195 countries are members. It does not conduct any research nor does it monitor climate-related data or parameters. Its remit does not focus on natural causes or trends of climate change. It is based at the WMO in Geneva.

Sources, residence and sinks

Relating these atmospheric concentrations to emissions, sources and sinks is a steadily evolving sphere of scientific inquiry. Certain inputs to the atmosphere can be discerned and readily quantified – carbon dioxide from fossil fuel burning* and CFCs from refrigerants for instance. Others, such as methane sources, are less certain, though about one-fifth of the methane emissions appear to be from fossil sources (coal seams, oil and natural gas, about 110 million tonnes per year).

* About 36.6 billion tonnes (9.98 GtC) from fossil fuels and cement production in 2018, plus about 5.5 Gt from land use change and deforestation (WMO Greenhouse Gas Bulletin #15).

Electricity generation is one of the major sources of carbon dioxide emissions, providing about one-third of the total and one-half of the increase expected 2005-30. Coal-fired generation* gives rise to twice as much carbon dioxide as natural gas per unit of power at the point of use, but hydro, nuclear power and most renewables do not directly contribute any. If all the world's nuclear power were replaced by coal-fired power, electricity's carbon dioxide emissions (now at least 11 billion tonnes per year) would rise by a quarter – about 3 billion tonnes per year. Conversely, there is scope for reducing coal's carbon dioxide contribution by substituting it for natural gas or nuclear, and by improving the efficiency of coal-fired generation itself, a process which is well under way. Substitution of coal by natural gas however requires consideration of methane leakage, and 3% leakage means that the global warming potential from using gas is the same as burning coal. In 2016 the Aliso Canyon underground gas storage in California was shut down after a massive leak of almost 100,000 tonnes of methane and over 7000 tonnes of ethane.

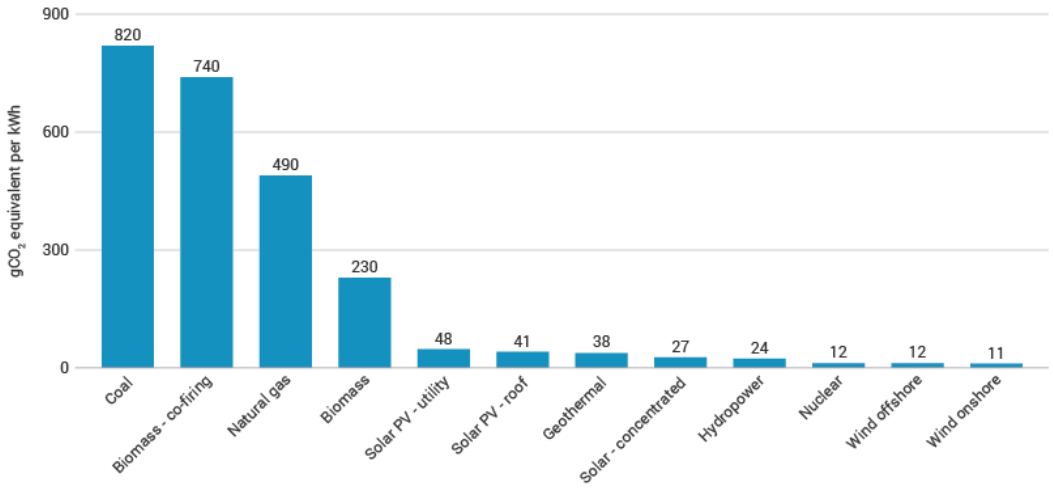
* in developed countries, with average 33% thermal efficiency. The difference is greater considering developing countries' average 25% efficiency.

Estimates of carbon dioxide concentrations in the atmosphere all show substantial increases. Global emissions of energy-related CO_2 are projected in several scenarios in the International Energy Agency's (IEA) annual *World Energy Outlook* reports.

Then there is the question of residence time in the atmosphere. For example methane has about an 11-year residence time before it is oxidised to carbon dioxide. Hydroxyl (OH) radicals are the main means of this oxidation. Carbon dioxide has a much longer residence time in the atmosphere, until it is either used up in photosynthesis or absorbed in rain or oceans.

Finally, in relating emissions to atmospheric concentrations, there is the question of sinks, or natural processes for breaking down or removing individual gases, particularly carbon dioxide. While the increase in carbon dioxide concentrations is remarkable, and the rate of anthropogenic emissions considerable (some 36 billion tonnes per year in 2014), even this is only about four percent of the natural flux between the atmosphere and the land and oceans. This perspective is important as a reminder that only a very small change to natural processes is required to compensate for (or exacerbate) anthropogenic emissions.

In fact, study of the atmospheric carbon cycle shows that less than half of the anthropogenic emissions show up as increased carbon dioxide levels. Both oceans and some terrestrial ecosystems provide sinks which function as a negative feedback, that is to say they have increased their uptake as the atmospheric concentration has increased. The IPCC summary in 2013 estimated that cumulative fossil fuel and cement production CO_2 emissions from 1750 to 2011 was about 365 GtC, with another 180 GtC from deforestation and land use change. Of this 545 GtC, about 240 GtC (44%) had accumulated in the atmosphere, 155 GtC (28%) had been taken up in the oceans with slight consequent acidification, and 150 GtC (28%) had accumulated in the terrestrial ecosystems. Ocean acidification – a decrease of about 0.03 in pH since 1990 – is an issue, possibly affecting organisms which rely on calcium carbonate. More recent data suggests that over the period 2014-2023, land and ocean sinks together continue to absorb around 56% of anthropogenic CO_2 emissions annually, meaning about 44% of emissions remain in the atmosphere.



Average life-cycle carbon dioxide-equivalent emissions for different electricity generators (Source: IPCC)

Where does the heat end up?

The focus of attention regarding global warming has been the atmosphere, where the heat is initially retained. However, more recently attention has turned to the oceans, whose thermal capacity is well over one hundred times that of the atmosphere. During recent decades many more measurements with higher accuracy have been made of temperatures in the upper layers of the ocean and in some parts of the deeper ocean. These have shown a slow but steady temperature rise broadly consistent with the increase in warming at the ocean's surface due to human influences, especially the release of greenhouse gases. Most of the net energy increase in the climate system in recent decades is stored in the oceans.

In the atmosphere, some warming of the troposphere is evident since the mid-20th century, and a temporary pause in warming over 1998-2012 was followed by more rapid warming. Recent studies show that the oceans lose heat to the atmosphere during warm El Niño events, while more heat penetrates to ocean depths in cold La Niñas. Such changes occur repeatedly over decades and more. In the major El Niño-Southern Oscillation event in 1997-98 the globally-averaged air temperature reached its highest level in the 20th century as the ocean lost heat to the atmosphere, mainly by evaporation, with a major effect on regional rainfall. Since then, the pause in tropospheric warming may be due to the timing of long Pacific and Atlantic ocean cycles.

Arctic sea ice is an indicator. Here there has been a significant decrease in sea ice since satellite records began in 1978. The

September minimum extent has decreased, and has the winter thickness. There is a positive feedback in summer since ice is reflective and open water absorbs heat. (In the Antarctic there has been no significant change in ice extent.)

Defining climate change prospects, effects and mitigation

The outcome of any significant climate change will be varied rather than simply an overall increase in average or nocturnal temperatures. Climate change is a global phenomenon, but manifests differently in different regions.

Climate researchers have designed models to predict the longer-term consequences both in air and ocean circulation patterns. These reproduce observed continental-scale surface temperature patterns and trends over many decades, including the more rapid warming since the mid-20th century and the cooling immediately following large volcanic eruptions, thus giving a range and probability of climatic impacts on different regions of the world. The models are constantly being refined, and in 2021 the IPCC reported that there is high confidence that climate models can now reproduce what has been observed globally and in most regions. Climate is defined as the statistical average of weather over a long period, typically 30 years.

IPCC Assessment Reports

The science behind the politics of climate change took a step forward and also ratcheted up concerns with the release of the Third Assessment Report from the UN's Intergovernmental Panel on Climate Change (IPCC), in 2001. The Fourth Assessment Report in 2007 further reduced uncertainties and led to calls for action. The Fifth Assessment Report in 2013-2014 repeated the call for a global agreement to limit carbon emissions, though it did slightly adjust downward the likely effects of increased CO_2 levels. The Sixth Assessment Report (AR6) in 2021 firmed up the scientific understanding, and said that the human influence on climate was unequivocal, and for the first time provided detailed regional assessments. Meanwhile there were two other relevant reports (see below).

Each IPCC Assessment Report is published in three parts. The first details the physical science basis for climate change. The second

covers the impacts of climate change, the options for adaptation and identifies where people and the environment are most vulnerable. The third part identifies options for mitigation of climate change.

A *Synthesis Report*, including a *Summary for Policymakers*, is also published for all three reports.

The first part of each successive report on the physical science basis (from Working Group I) has concluded that the evidence that human-derived greenhouse gas emissions have already had an impact on the climate has strengthened over time. Furthermore, confidence has grown in predictions of the impacts of future greenhouse gas emissions.

The first two of four headline statements from *Climate Change 2014: Synthesis Report* (the Synthesis Report of the Fifth Assessment Report (AR5, 2013)) are:

- Human influence on the climate system is clear, and recent anthropogenic emissions of greenhouse gases are the highest in history. Recent climate changes have had widespread impacts on human and natural systems.

- Continued emission of greenhouse gases will cause further warming and long-lasting changes in all components of the climate system, increasing the likelihood of severe, pervasive and irreversible impacts for people and ecosystems. Limiting climate change would require substantial and sustained reductions in greenhouse gas emissions which, together with adaptation, can limit climate change risks.

Among the AR5 findings on physical science were:

- More than half of the observed increase in globally averaged temperatures since the mid-20th century is *extremely likely* (95%+ probability) due to human influence, notably the observed increase in anthropogenic greenhouse gas concentrations.

- Greenhouse gases contributed a global mean surface warming *likely* (66%+ confidence) to be in the range of 0.5 °C to 1.3 °C over the period 1951-2010.

- More than 60% of the net energy increase in the climate system was stored in the upper ocean (0-700 m) from 1971 to 2010,

and about 30% is stored in the ocean below 700 m.

- Anthropogenic influences likely contributed to the retreat of glaciers since the 1960s and to the diminution of the Greenland ice sheet since 1993.

- Multiple lines of evidence support very substantial Arctic warming since the mid-20th century, and anthropogenic influences have *very likely* contributed to Arctic sea ice loss since 1979.

- Global average sea level rose at an average rate of 2.0 mm per year over 1971 to 2010. The rate was faster over 1993 to 2010, about 3.2 mm per year.

- It is *very likely* that there is a substantial anthropogenic contribution to the global mean sea level rise since the 1970s. This is based on the *high confidence* in an anthropogenic influence on the two largest contributions to sea level rise – thermal expansion and glacier mass loss.

- More intense and longer droughts have been observed over wider areas since the 1970s, particularly in the tropics and subtropics.

- Widespread changes in extreme temperatures have been observed over the last 50 years. Cold days, cold nights and frost have become less frequent, while hot days, hot nights, and heat waves have become more frequent

- The global atmospheric concentration of methane has increased from a pre-industrial value of about 715 ppb to 1820 ppb in 2011.

- The combined radiative forcing due to increases in carbon dioxide, methane, nitrous oxide and halocarbons is +2.83 W/m^2, and its rate of increase during the industrial era is *very likely* to have been unprecedented in more than 10,000 years.

In AR5, four scenarios for future carbon emissions to 2100 ranged from means of 270 GtC, assuming substantial cuts in emissions and correlated with best-case radiative forcing of 2.5 W/m^2, to 1685 GtC correlated with 8.5 W/m^2 radiative forcing. Accordingly, it predicted that, based on the range of scenarios, by the end of the 21st century:

- Global surface temperature change *likely* to exceed 1.5 °C relative to 1850 to 1900 for two scenarios, be about 2 °C in one, and approach 4 °C in the other.

- A sea level rise most likely to be 47-63 cm, due more to thermal expansion than retreating glaciers and Greenland ice cap.

- Arctic summer sea ice disappearing in second half of the century in all but the lowest scenario.

- It is *virtually certain* that there will be more frequent hot and fewer cold temperature extremes over most land areas on daily and seasonal timescales as global mean temperatures increase. It is *very likely* that heat waves will occur with a higher frequency and duration. Occasional cold winter extremes will continue to occur.

- It is *virtually certain* that near-surface permafrost extent at high northern latitudes will be reduced as global mean surface temperature increases.

- Most aspects of climate change will persist for many centuries even if emissions of CO_2 are stopped. This represents a substantial multi-century climate change commitment created by past, present and future emissions of CO_2.

There remains considerable uncertainty regarding the above effects on the frequency and intensity of hurricanes, tornadoes and to some extent, droughts.

Among the Sixth Assessment Report (AR6, 2021)* findings on physical science were:

- Global surface temperature has increased by 1.09 [0.95-1.20] °C from the 1850-1900 baseline to 2011-2020, and the last decade was *more likely than not* warmer than any multi-centennial period after the last interglacial, roughly 125,000 years ago. The *likely* range of human-induced change in this is 0.8-1.3 °C, with a central estimate of 1.07 °C. Thus warming is overwhelmingly due to human influence. There has been negligible long-term influence from solar activities and volcanoes.

- The anthropogenic effective radiative forcing due to increases in carbon dioxide, methane, nitrous oxide and halocarbons and

partly compensated by negative aerosol effect, is +2.72 W/m^2 (*medium confidence*).

- Over the past four to six decades, it is virtually certain that the global ocean has warmed, with human influence extremely likely the main driver since the 1970s, making climate change irreversible over centuries to millennia (*medium confidence*). About 90% of the net increase in global energy inventory is in the oceans.

- It is *virtually certain* that upper ocean stratification has increased since 1970 and that seawater pH has declined globally over the last 40 years, and that the main driver is uptake of anthropogenic CO_2.

- Human influence was *very likely* the main driver of observed reductions in Arctic sea ice since the late 1970s (with late-summer sea ice loss likely unprecedented for at least 1000 years) and the widespread retreat of glaciers (unprecedented in at least the last 2000 years, *medium confidence*).

- By contrast, Antarctic sea ice area experienced no significant net change since 1979, and there is only *low confidence* in its projected changes.

- It is *likely* that the Antarctic ice sheet has lost substantial mass since 1979, contributing at least 6 mm to the global mean sea level rise since 1992.

- It is *virtually certain* that the Greenland ice sheet has lost mass since the 1990s, with human influence a contributing factor (*medium confidence*).

- Global mean sea level rose by 200 mm (150-250 mm) since 1901 at an average rate of 1.7 mm/yr, but the rate of rise accelerated to 3.7 mm/yr from 2006 (*high confidence*) due to ocean warming and melting glaciers. Human activity was very likely the cause from 1970.

- Paleo-evidence shows that global mean sea level has been about 70 metres higher and 130 metres lower than present within the past 55 million years and was *likely* 5-10 metres higher during the last interglacial.

- Human-induced climate change has contributed to increases in agricultural and ecological droughts in some regions due to

increases in evapotranspiration (*medium confidence*).

- There is *high confidence* that climate change has driven changes in the global water cycle since the mid-20th century.

- It is *likely* that the proportion of intense tropical cyclones has increased over the last four decades and that this cannot be explained entirely by natural variability. There is *low confidence* in observed recent changes in the total number of extratropical cyclones.

- It is *virtually certain* that the frequency and intensity of hot extremes and the intensity and duration of heatwaves have increased since 1950, but with regional variability. The frequency and intensity of heavy precipitation events have increased over a majority of those land regions with good observational coverage (*high confidence*).

- Over the past half century, key aspects of the biosphere have changed in ways that are consistent with large-scale warming: climate zones have shifted poleward, and the growing season length in the Northern Hemisphere extratropics has increased (*high confidence*). There has been increasing productivity of the land biosphere due to the increasing atmospheric CO_2 concentration as the main driver (*medium confidence*). Global-scale vegetation greenness has increased since the 1980s (*high confidence*).

- There is *high confidence* that climate models can now reproduce the recent observed mean state and overall warming of temperature extremes both globally and in most regions.

Scenarios in AR6 cover a broader range of emissions futures than AR5, with less uncertainty due to better understanding of climate drivers and feedbacks. They include high CO_2 emissions scenarios without climate change mitigation as well as a low CO_2 emissions scenario reaching net zero CO_2 emissions around mid-century. In this report, a core set of five illustrative 'Shared Socioeconomic Pathways' (SSPs) is used to explore climate change over the 21st century and beyond. They are labelled SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5, and span a wide range of radiative forcing levels in 2100 – *i.e.* 1.9-8.5 W/m^2. The first could lead to warming below 1.5 °C in 2100. The near-linear relationship between cumulative CO_2 emissions and maximum global surface temperature increase caused by CO_2 implies that stabilizing human-induced global temperature increase at any level requires

net anthropogenic CO_2 emissions to become zero (high confidence, TS 3.3.1).

The likelihood of individual scenarios is not part of AR6, but the report notes that the very high emissions and warming scenario SSP5-8.5 "has been debated in light of recent developments in the energy sector" and discounted but cannot be entirely ruled out. It projects a very great increase in coal use and has been carried forward from earlier modelling without real modification. Including this improbable, obsolete and extreme scenario, it is predicted that, based on the range of scenarios, by the end of the 21st century climate change would result in the following:

- Compared with the 1850-1900 baseline, average global surface temperature over the period 2081-2100 is *very likely* to be higher by 1.0-1.8 °C in the low CO_2 emissions scenario SSP1-1.9 and by 3.3-5.7 °C in the high CO_2 emissions scenario SSP5-8.5.

- SSP1-2.6 and SSP2-4.5 represent stronger climate change mitigation and lower emissions, the first limiting warming to 2 °C.

- During the near-term (by 2040), a 1.5 °C increase in global surface temperature, relative to 1850-1900, is *very likely* to occur in scenario SSP5-8.5, *likely* to occur in scenarios SSP2-4.5 and SSP3-7.0, and *more likely than not* to occur in scenarios SSP1-1.9 and SSP1-2.6.

- For the mid-term period 2041-2060, the 2 °C global warming level is *very likely* to be crossed under SSP5-8.5, *likely* to be crossed under SSP3-7.0, and *more likely than not* to be crossed under SSP2-4.5.

- The 2 °C global warming level is *unlikely* to be crossed under SSP1-2.6 and *extremely unlikely* to be crossed under SSP1-1.9 any time this century.

- The amount of ocean warming observed since 1971 will *likely* at least double by 2100 under a low warming scenario (SSP1-2.6) and increase by 4-8 times under a high warming scenario (SSP5-8.5).

- Mass loss over the 21st century is *virtually certain* for the Greenland ice sheet and *likely* for the Antarctic ice sheet.

- Global mean sea level is projected to rise by 0.28-0.55 m (*likely* range) under SSP1-1.9 and 0.63-1.02 m (*likely* range) under SSP5-8.5 relative to the 1995-2014 average (*medium confidence*).

- Arctic late summer sea ice will disappear by the end of the century in the high emissions scenarios.

- Near-surface permafrost at high northern latitudes will be reduced as global mean surface temperature increases (*high confidence*).

- There is *medium confidence* that the projected *very likely* decline in the Atlantic Meridional Overturning Circulation (AMOC) will not involve an abrupt collapse before 2100, though such an event would have significant consequences.

- It is *virtually certain* that the frequency and intensity of hot extremes and the intensity and duration of heatwaves will further increase even if global warming is stabilized at 1.5 °C.

- It is *extremely likely* that the frequency and intensity of heavy precipitation events will increase over most continents with additional global warming.

- Over the 21st century, the total land area subject to drought will increase and droughts will become more frequent and severe (*high confidence*).

- There is *high confidence* that climate change will increase the variability of the global water cycle in most regions under all scenarios.

- More regions are affected by increases in agricultural and ecological droughts with increasing global warming (*high confidence*).

- For global climate indicators, evidence for abrupt change is limited, but deep ocean warming, acidification and sea level rise are committed to ongoing change for millennia after global surface temperatures initially stabilize and are irreversible on human timescales (*very high confidence*).

In the AR6 Technical Summary, the following terms have been used to indicate the assessed likelihood of an outcome or a result: *virtually certain* 99-100% probability, *very likely* 90-100%, *likely* 66-100%, *about as likely as*

not 33-66%, *unlikely* 0-33%, *very unlikely* 0-10%, *exceptionally unlikely* 0-1%. The level of confidence in validity is expressed using five qualifiers: *very low*, *low*, *medium*, *high*, and *very high*.

The second part (Working Group II contribution) of each IPCC Assessment Report deals with impacts, adaptation and vulnerabilities. The third part (Working Group III contribution) deals with the mitigation of climate change, outlining the prospects and options for change, particularly in the energy sector, which accounts for 60% of emissions. The second and third parts of AR6 have not yet been released.

The second parts of successive reports have concluded that climate change will have significant impacts, including increased stress on water supplies and a widening threat of species extinction.

The third parts of successive reports have each agreed that major changes are required to adopt low-carbon energy technologies, and that a single global carbon price is key to achieving emissions reductions.

The Working Group III contribution to AR5 stated: "Scenarios reaching atmospheric concentration levels of about 450 to about 500 ppm CO_2eq by 2100 are characterized by a tripling to nearly a quadrupling of the global share of zero- and low-carbon energy supply from renewables, nuclear energy, fossil energy with carbon dioxide capture and storage (CCS), and bioenergy with CCS (BECCS), by the year 2050 relative to 2010 (about 17%). The increase in total global low-carbon energy supply is from three-fold to seven-fold over this same period. Many models could not reach 2100 concentration levels of about 450 ppm CO_2eq if the full suite of low-carbon technologies is not available."

Other science-based reports

In December 2011 a report from the Global Carbon Project (GCP), a research consortium, pointed out that in 2010 CO_2 emissions from fossil fuels and cement production were 33.4 ±1.8 Gt CO_2, 41% of which was from coal and 34% from oil. These emissions were the highest in human history and 49% higher than in 1990 (the Kyoto reference year). Coal burning was responsible for 52% of the fossil fuel emissions growth in 2010 (gas 23% and liquid 18%). CO_2 emissions from deforestation and other land use change were 3.3 ±2.6 Gt CO_2 (0.9 ±0.7 GtC) in 2010, leading to total emissions (including fossil fuel and land use change) of 36.7 ±3.3 Gt CO_2.

According to the GCP these ended up 50% in the atmosphere, 26% in biomass and 24% in oceans.

The National Oceanic and Atmospheric Administration (NOAA) Annual Greenhouse Gas Index (AGGI) in 2017 showed that from 1990 to 2016, radiative forcing by long-lived greenhouse gases increased by 40%, with CO_2 accounting for about 80% of this increase.

The IPCC prepared a special report on Global Warming of 1.5 °C, and how this might be achieved in the context of sustainable development and efforts to eradicate poverty. The first draft cited about 3000 publications, two-thirds of them being since the Fifth Assessment Report. It was released in October 2018 and said:

- Human-induced warming reached approximately 1 °C (*likely* between 0.8 °C and 1.2 °C) above pre-industrial levels in 2017, increasing at 0.2 °C (*likely* between 0.1 °C and 0.3 °C) per decade (*high confidence*).
- Past emissions alone are unlikely to raise global-mean temperatures to 1.5 °C above pre-industrial levels (*medium confidence*), but past emissions do commit to other changes, such as further sea level rise (*high confidence*).
- Limiting warming to 1.5 °C is not physically impossible but would require unprecedented transitions in all aspects of society.
- There are clear benefits to keeping warming to 1.5 °C compared with 2 °C or higher.
- Limiting warming to 1.5 °C can go hand-in-hand with reaching other world goals such as achieving sustainable development and eradicating poverty.

An IPCC report from Working Groups I and II (physical science & impacts/adaptation) was released in September 2019, on *The Ocean and Cryosphere in a Changing Climate*. Regarding the basic science, it said:

- Over the last decades, global warming has led to widespread shrinking of the cryosphere, with mass loss from ice sheets and glaciers (*very high confidence*), reductions in snow cover (*high confidence*) and Arctic sea ice extent and thickness (*very high*

confidence), and increased permafrost temperature (*very high confidence*).

- It is *virtually certain* that the global ocean has warmed unabated since 1970 and has taken up more than 90% of the excess heat in the climate system (*high confidence*). Since 1993, the rate of ocean warming has more than doubled (*likely*). ... By absorbing more CO_2, the ocean has undergone increasing surface acidification (*virtually certain*).

- Global mean sea level (GMSL) is rising, with acceleration in recent decades due to increasing rates of ice loss from the Greenland and Antarctic ice sheets (*very high confidence*), as well as continued glacier mass loss and ocean thermal expansion.

- Ecosystems in high mountain and polar regions and also marine ecosystems have changed (*high confidence*).

A 'high-level synthesis report', *United in Science*, compiled by the World Meteorological Organization (WMO) with UNEP and others for the Science Advisory Group of the UN Climate Action Summit in 2019 added to the IPCC *Ocean & Cryosphere* report, including:

- The average global temperature for 2015-2019 is on track to be the warmest of any equivalent period on record. It is currently estimated to be 1.1 °C (±0.1 °C) above pre-industrial (1850-1900) times.

- CO_2 emissions from fossil fuel use continue to grow by over 1% annually and 2% in 2018, having reached a new high of 53.5 Gt/yr CO_2 equivalent in 2017.

- The ocean absorbs nearly 25% of the annual emissions of anthropogenic CO_2 thereby helping to alleviate the impacts of climate change on the planet. The absorbed CO_2 reacts with seawater and increases the acidity of the ocean.

- Current levels of CO_2, CH_4 and N_2O represent 146%, 257% and 122% respectively of pre-industrial levels (pre-1750).

- Implementing current unconditional nationally-determined contributions (NDC) to reducing CO_2 emissions under the Paris Agreement would lead to a global mean temperature rise between 2.9 °C and 3.4 °C by 2100 relative to pre-industrial levels, and continuing thereafter.

- The current level of NDC ambition needs to be roughly tripled for emissions reduction to be in line with the 2 °C goal and increased five-fold for the 1.5 °C goal.

- Meeting the Paris Agreement requires immediate and all-inclusive action encompassing deep decarbonization complemented by ambitious policy measures, protection and enhancement of carbon sinks and biodiversity, and effort to remove CO_2 from the atmosphere.

In March 2023 the IPCC released its Synthesis Report (AR6), providing a fully integrated assessment of science, impacts and mitigation and emphasizing that deep, rapid and sustained emissions reductions this decade are critical to limit warming to 1.5 °C.

Tipping point?

The joint February 2014 report by the UK Royal Society and the US National Academy of Sciences, *Climate Change: Evidence & Causes*, presents a lot of information, including that from the IPCC Fifth Assessment Report, as above. It also says: "Results from the best available climate models do not predict abrupt changes in such systems (often referred to as tipping points) in the near future. However, as warming increases, the possibilities of major abrupt change cannot be ruled out." However, "the climate system involves many competing processes that could switch the climate into a different state once a threshold has been exceeded.

"A well-known example is the south-north ocean overturning circulation, which is maintained by cold salty water sinking in the North Atlantic and which involves the transport of extra heat to the North Atlantic via the Gulf Stream. During the last ice age, pulses of freshwater from the ice sheet over North America led to slowing down of this overturning circulation and to widespread changes in climate around the Northern Hemisphere. Freshening of the North Atlantic from the melting of the Greenland ice sheet is however, much less intense and hence is not expected to cause abrupt changes. As another example, Arctic warming could destabilise methane (a greenhouse gas) trapped in ocean sediments and permafrost, potentially leading to a rapid release of a large amount of methane. If such a rapid release occurred, then major, fast climate changes would ensue.

“Such high-risk changes are considered unlikely in this century, but are by definition hard to predict.”

The WMO 2019 *United in Science report* said: “With continued warming, systems can reach tipping points where they rapidly collapse or a major, largely unstoppable transformation is initiated. Scientists have studied plausible pathways to a ‘Hothouse Earth’ scenario, where interacting tipping points could potentially lead to a cascading effect where Earth’s temperature heats up to a catastrophic 4-5 °C. Another study estimates that unmitigated emissions could reverse a multimillion-year cooling trend in less than two centuries.”

Geological context and perspective

The Earth's climate has changed over millions of years, and there have been times when CO_2 levels were higher than today. Evidence for climate change is preserved in a wide range of geological settings, including marine and lake sediments, ice sheets, fossil corals, stalagmites and fossil tree rings. The following information comes from a 2010 position statement from the Geological Society of London.

The Earth’s climate has been gradually cooling for most of the last 50 million years. At the beginning of that cooling (in the early Eocene), the global average temperature was about 6-7 °C warmer than now. About 34 million years ago, at the end of the Eocene, ice caps coalesced to form a continental ice sheet on Antarctica. In the northern hemisphere, as global cooling continued, local ice caps and mountain glaciers gave way to large ice sheets around 2.6 million years ago.

Over the past 2.6 million years (the Pleistocene and Holocene), the Earth’s climate has been on average cooler than today, and often much colder. That period is known as the ‘Ice Age’, a series of glacial episodes separated by short warm ‘interglacial’ periods that lasted between 10,000-30,000 years. We are currently living through one of these interglacial periods. The present warm period (known as the Holocene) became established only 11,500 years ago, since when our climate has been relatively stable. Although we currently lack the large Northern Hemisphere ice sheets of the Pleistocene, there are of course still large ice sheets on Greenland and Antarctica.

Global sea level is very sensitive to changes in global temperatures. Ice sheets grow when the Earth cools and melt when it warms.

Warming also heats the ocean, causing the water to expand and the sea level to rise. When ice sheets were at a maximum during the Pleistocene, world sea level fell to at least 120 metres below where it stands today. Relatively small increases in global temperature in the past have led to sea level rises of several metres. During parts of the previous interglacial period, when polar temperatures reached 3-5 °C above today's, global sea levels were higher than today's by around 4-9 metres.

Relatively rapid global warming has occurred in the past. About 55 million years ago, at the end of the Paleocene, there was a sudden warming event in which temperatures rose by about 6 °C globally and by 10-20 °C at the poles. Carbon isotopic data show that this warming event (called by some the Paleocene-Eocene Thermal Maximum, or PETM) was accompanied by a major release of 1500-2000 billion tonnes or more of carbon (5550-7400 billion tonnes or more of CO_2) into the ocean and atmosphere. This injection of carbon may have come mainly from the breakdown of methane hydrates beneath the deep sea floor, perhaps triggered by volcanic activity superimposed on an underlying gradual global warming trend that peaked some 50 million years ago in the early Eocene. CO_2 levels were already high at the time. It took the Earth's climate around 100,000 years or more to recover, showing that a CO_2 release of such magnitude may affect the Earth's climate for that length of time.

Recent estimates suggest that at times between 5.2 and 2.6 million years ago (during the Pliocene), the carbon dioxide concentrations in the atmosphere reached between 330 and 400 ppm. During those periods, global temperatures were 2-3 °C higher than now, and sea levels were higher than now by 10-25 metres, implying that global ice volume was much less than today. There were large fluctuations in ice cover on Greenland and western Antarctica during the Pliocene, and during the warm intervals those areas were probably largely free of ice.

In 2013 the Geological Society published an addendum to its 2010 position statement, which said that new climate data from the geological record strengthen the 2010 statement's original conclusion that CO_2 is a major modifier of the climate system, and that human activities are responsible for recent warming. Geologists have recently contributed to improved estimates of climate sensitivity (defined as the increase in global mean temperature resulting from a doubling in atmospheric CO_2 levels). Studies of the Last Glacial Maximum (about 20,000 years ago) suggest that the climate sensitivity, based on rapidly-acting factors like snow melt, ice melt and the behaviour of clouds and water

vapour, lies in the range 1.5-6.4 °C. Furthermore, slow-acting factors like the decay of large ice sheets and the operation of the full carbon cycle, suggest that this could double the climate sensitivity.

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